JEE Mains 2020 Jan Chapter wise Question Bank

Matrices and Determinants

Q1

If α is a roots of equation $x^2 + x + 1 = 0$ and $A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}$ then A^{31} equal to :

यदि α समीकरण $x^2 + x + 1 = 0$ का एक मूल है तथा $A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}$ तब A^{31} बराबर है–

(1) A (2) A^2 (3) A^3 (4) A^4

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Sol

(3)

$A^2 = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

$\Rightarrow A^4 = I$

$\Rightarrow A^{30} = A^{28} \times A^3 = A^3$

02

If system of equations

$2x + 2ay + az = 0$

$2x + 3by + bz = 0$

$2x + 4cy + cz = 0$ have non-trivial solution

then

(1) $a + b + c = 0$

(2) a,b,c are in A.P.

(3) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

(4) a, b, c in G.P.

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Sol

For non-trivial solution

अशून्य हल के लिए

$$\begin{vmatrix} 2 & 2a & a \\ 2 & 3b & b \\ 2 & 4c & c \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & 2a & a \\ 1 & 3b & b \\ 1 & 4c & c \end{vmatrix} = 0$$

$(3bc - 4bc) - (2ac - 4ac) + (2ab - 3ab) = 0$

$-bc + 2ac - ab = 0$

$ab + bc = 2ac$

a, b, c in H.P.

$\Rightarrow \quad \frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ in A.P.

03

Let $A = [a_{ij}]$, $B=[b_{ij}]$ are two 3×3 matrices such that $b_{ij} = \lambda^{i+j-2} a_{ij}$ & $|B| = 81$. Find $|A|$ if $\lambda = 3$.

माना $A = [a_{ij}]$, $B=[b_{ij}]$, 3×3 क्रम की दो आव्यूह इस प्रकार है कि $b_{ij} = \lambda^{i+j-2} a_{ij}$ तथा $|B| = 81$. यदि $\lambda = 3$ है तब $|A|$का मान है–

(1) $\frac{1}{9}$ (2) 3 (3) $\frac{1}{81}$ (4) $\frac{1}{27}$

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Sol

(1)

$$|B| = \begin{vmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{vmatrix} = \begin{vmatrix} 3^0 a_{11} & 3^1 a_{12} & 3^2 a_{13} \\ 3^1 a_{21} & 3^2 a_{22} & 3^3 a_{23} \\ 3^2 a_{31} & 3^3 a_{32} & 3^4 a_{33} \end{vmatrix} \Rightarrow 81 = 3^3.3.\,3^2 |A| \Rightarrow 3^4 = 3^6 |A| \Rightarrow |A| = \frac{1}{9}$$

04

If system of equation यदि समीकरणों का निकाय

$x + y + z = 6$

$x + 2y + 3z = 10$

$3x + 2y + \lambda z = \mu$

has more than two solutions. Find $(\mu - \lambda^2)$

के दो से अधिक हल है तब $(\mu - \lambda^2)$ का मान होगा–

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Sol

13

$x + y + z = 6$ .... (1)

$x + 2y + 3z = 10$ .... (2)

$3x + 2y + \lambda z = \mu$ .... (3)

from (1) and तथा (2) से

if $z = 0 \Rightarrow x + y = 6$ and और $x + 2y = 10$

$\Rightarrow y = 4, x = 2$

(2, 4, 0)

if $y = 0 \Rightarrow x + z = 6$ and और $x + 3z = 10$

$\Rightarrow z = 2$ and और $x = 4$

(4, 0, 2)

so इसलिए $3x + 2y + \lambda z = \mu$

must pass through गुजरती है (2, 4, 0) and तथा (4, 0, 2)

so इसलिए$6 + 8 = \mu \Rightarrow \mu = 14$

and तथा $12 + 2\lambda = \mu$

$12 + 2\lambda = 14 \Rightarrow \lambda = 1$

so इसलिए $\mu - \lambda^2 = 14 - 1$

$= 13$

05

A is a 3×3 matrix whose elements are from the set $\{-1, 0, 1\}$. Find the number of matrices A such that $tr(AA^T) = 3$. Where tr(A) is sum of diagonal elements of matrix A.

A एक 3×3 क्रम का आव्यूह है, जिसके अवयव समुच्चय $\{-1, 0, 1\}$ से लिए गये है। तब आव्यूह A की संख्या ज्ञात करो जो इस प्रकार है कि $tr(AA^T) = 3$ जहाँ tr(A) आव्यूह A के विकर्ण के अवयवों का योग है।

(1) 572 (2) 612 (3) 672 (4) 682

8th Jan Morning

Sol

(3)

Let माना $A = [a_{ij}]_{3\times3}$

$tr(AA^T) = 3$

$a_{11}^2 + a_{12}^2 + a_{13}^2 + a_{21}^2 + \ldots\ldots\ldots + a_{33}^2 = 3$

possible cases

संभावित स्थितियाँ

$$\left.\begin{array}{ll} 0,0,0,0,0,0,1,1,1 & \to 1 \\ 0,0,0,0,0,0,-1,-1,-1 & \to 1 \\ 0,0,0,0,0,0,1,1,-1 & \to 3 \\ 0,0,0,0,0,0,-1,1,-1 & \to 3 \end{array}\right\} {}^9C_6 \times 8 = 84 \times 8 = 672$$

06

The system of equation $3x + 4y + 5z = \mu$
$x + 2y + 3z = 1$
$4x + 4y + 4z = \delta$
is inconsistent, then (δ, μ) can be

(1) (4, 6) (2) (3, 4) (3) (4, 3) (4) (1, 0)

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Sol

(3)

Note $D = \begin{vmatrix} 3 & 4 & 5 \\ 1 & 2 & 3 \\ 4 & 4 & 4 \end{vmatrix}$ $(R_3 \to R_3 - 2R_1 + 3R_2)$

$= \begin{vmatrix} 3 & 4 & 5 \\ 1 & 2 & 3 \\ 0 & 0 & 0 \end{vmatrix} = 0$

Now let $P_3 \equiv 4x + 4y + 4z - \delta = 0$. If the system has solutions it will have infinite solution,
so $P_3 \equiv \alpha P_1 + \beta P_2$
Hence $3\alpha + \beta = 4$ & $4\alpha + 2\beta = 4 \Rightarrow \alpha = 2$ & $\beta = -2$
So for infinite solution $2\mu - 2 = \delta \Rightarrow$ for $2\mu \neq \delta + 2$
system inconsistent

07

Let $A = \begin{bmatrix} 2 & 2 \\ 9 & 4 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ then value of $10\,A^{-1}$ is –

माना $A = \begin{bmatrix} 2 & 2 \\ 9 & 4 \end{bmatrix}$ तथा $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ तो $10\,A^{-1}$ का मान होगा :

(1) $4I - A$ (2) $6I - A$ (3) $A - 4I$ (4) $A - 6I$

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Sol

(4)

Characteristics equation of matrix 'A' is
आव्यूह 'A' की अभिलाक्षणिक समीकरण है –

$\begin{vmatrix} 2-x & 2 \\ 9 & 4-x \end{vmatrix} = 0 \quad \Rightarrow \quad x^2 - 6x - 10 = 0$

$\therefore \quad A^2 - 6A - 10\,I = 0$

$\Rightarrow \quad 10A^{-1} = A - 6I$

08

$\lambda x + 2y + 2z = 5$
$2\lambda x + 3y + 5z = 8$
$4x + \lambda y + 6z = 10$
for the system of equation check the correct option.

(1) Infinite solutions when $\lambda = 8$
(2) Infinite solutions when $\lambda = 2$
(3) no solutions when $\lambda = 8$
(4) no solutions when $\lambda = 2$

8^{th} Jan Evening

Sol

(4)

$$D = \begin{vmatrix} \lambda & 2 & 2 \\ 2\lambda & 3 & 5 \\ 4 & \lambda & 6 \end{vmatrix}$$

$D = (\lambda + 8)(2 - \lambda)$
for $\lambda = 2$

$$D_1 = \begin{vmatrix} 5 & 2 & 2 \\ 8 & 3 & 5 \\ 10 & 2 & 6 \end{vmatrix}$$

$= 5[18 - 10] - 2[48 - 50] + 2(16 - 30]$
$= 40 + 4 - 28 \neq 0$
No solutions for $\lambda = 2$

09

If plane
$x + 4y - 2z = 1$
$x + 7y - 5z = \beta$
$x + 5y + \alpha z = 5$
intersects in a line $(R \times R \times R)$ then $\alpha + \beta$ is equal to
यदि समतल
$x + 4y - 2z = 1$
$x + 7y - 5z = \beta$
$x + 5y + \alpha z = 5$
एक रेखा में $(R \times R \times R)$ तो $\alpha + \beta$ बराबर है–

(1) 0 (2) 10 (3) – 10 (4) 2

9^{th} Jan Morning

Sol

(2)

$$\Delta = 0 \Rightarrow \begin{vmatrix} 1 & 4 & -2 \\ 1 & 7 & -5 \\ 1 & 5 & \alpha \end{vmatrix} = 0$$

$(7\alpha + 25) - (4\alpha + 10) + (-20 + 14) = 0$

$3\alpha + 9 = 0 \Rightarrow \alpha = -3$

Also तथा $D_z = 0 \Rightarrow \begin{vmatrix} 1 & 4 & 1 \\ 1 & 7 & \beta \\ 1 & 5 & 5 \end{vmatrix} = 0$

$1(35 - 5\beta) - (15) + 1\ (4\beta - 7) = 0$

$\beta = 13$

10

If $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{bmatrix}$, $B = \text{adj}(A)$ and $C = 3A$ then $\dfrac{|\text{adj}B|}{|C|}$ is

यदि $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{bmatrix}$, $B = \text{adj}(A)$ तथा $C = 3A$ तब $\dfrac{|\text{adj}B|}{|C|}$ बराबर है–

(1) 8 (2) 4 (3) 2 (4) 16

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Sol

(1)

$$|A| = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{vmatrix} = ((9+4) - 1(3-4) + 2(-1-3))$$

$= 13 + 1 - 8 = 6$

$|\text{adj}B| = |\text{adjadj}A| = |A|^{(n-1)^2} = |A|^4 = (36)^2$

$|C| = |BA| = 3^3 \times 6$

$$\frac{|\text{adj}B|}{|C|} = \frac{36 \times 36}{3^3 \times 6} = 8$$

11

If यदि $f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$ and तथा $a - 2b + c = 1$ then तब

(1) $f(50) = 1$
(2) $f(-50) = -1$
(3) $f(50) = 501$
(4) $f(50) = -501$

9th Jan Evening

Sol

(1)

Apply $R_1 = R_1 + R_3 - 2R_2$ प्रयोग करने पर

$$\Rightarrow f(x) = \begin{vmatrix} 1 & 0 & 0 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix} \quad \Rightarrow f(x) = 1 \quad \Rightarrow f(50) = 1$$

12

If यदि $7x + 6y - 2z = 0$
$3x + 4y + 2z = 0$
$x - 2y - 6z = 0$ then which option is correct
$x - 2y - 6z = 0$ तब निम्न में से कौनसा विकल्प सत्य है–

(1) no. solution
(2) only trivial solution
(3) Infinite non trivial solution for $x = 2z$
(4) Infinite non trivial solution for $y = 2z$

(1) कोई हल नही
(2) केवल तुच्छ हल
(3) $x = 2z$ के लिए अनन्त अतुच्छ हल
(4) $y = 2z$ के लिये अनन्त अतुच्छ हल

9th Jan Evening

Sol

(3)

$$\begin{vmatrix} 7 & 6 & -2 \\ 3 & 4 & 2 \\ 1 & -2 & -6 \end{vmatrix}$$

$= 7(-20) - 6(-20) - 2(-10)$
$= -140 + 120 + 20 = 0$
so infinite non-trivial solution exist
अतः अनन्त अतुच्छ हल का अस्तित्व होगा
now equation (1) + 3 equation (3)
अब समीकरण (1) + 3 समीकरण (3)
$10x - 20z = 0$
$x = 2z$